

Course Syllabus

 Edit

CIST 1423 – Virtual Reality Programming and Technology

Course Detail

Title: Virtual Reality Programming and Technology

Prefix and Number: CIST 1423

Term and Section CRN: Fall 2026, CRN 25461

Upper-Level Course Requirements: Completion of ENG 0101, ENG 0102, FS 0104 (or FS 0102), and the MATH competency.

Class Meeting Days, Times, and Location

Wednesdays, 6:00 PM – VR Lab, Duke Engineering and Information Technology Building, Room 113

Instructor Information

Jeremy Callinan

Office Location: VR Lab, Duke Engineering and Information Technology Building, Room 113

University Email: jpcst52@pitt.edu (alternate: jeremy.callinan@gmail.com)

Campus Phone Number: 814-758-4206

Office Hours: Wednesdays, 8:45–10:00 AM, on site and online

Course Description

Students learn modern game and VR development using Unity, GitHub, and C#, gaining practical and theoretical knowledge through workflows that mirror professional software development. Assignments are hands-on, with students creating small Unity prototypes (2D, 3D, and VR) before scaling up to a final project. Students will also help staff the VR Lab for campus events (e.g., Alumni Weekend), operating VR experiences on the Vive Pro 2 (5K resolution, wireless adapter) for visitors. By combining project-based learning with structured skill development, students leave the course confident in their ability to design, script, test, and publish immersive experiences.

Course Student Learning Outcomes

By the end of this course, students will be able to:

- Explain core VR hardware and interaction concepts (stereo rendering, depth perception, sensors, controllers) across platforms such as Oculus Quest 2, HTC Vive Pro 2, and Microsoft HoloLens.

- Build Unity scenes using GameObjects, prefabs, physics, colliders, and C# scripting.
- Implement VR interactions using the XR Interaction Toolkit, including locomotion and object manipulation.
- Use GitHub for version control and collaborative development, including branching and pull requests.
- Design, build, test, and present an original virtual environment as a final project.

Required and Recommended Course Materials

XR Development with Unity: A Beginner's Guide to Creating Virtual, Augmented, and Mixed Reality Experiences Using Unity, Anna Braun and Raffael Rizzo (2023, 1st ed.). Available via the [Pitt Library](https://pitt.primo.exlibrisgroup.com/permalink/01PITT_INST/u9csjr/cdi_walterdegruyter_marc_9781805120049?context=PC&vid=01PITT_INST&lang=en&adocid=cdi_walterdegruyter_marc_9781805120049&context=PC&vid=01PITT_INST&lang=en&ad)  (https://pitt.primo.exlibrisgroup.com/permalink/01PITT_INST/u9csjr/cdi_walterdegruyter_marc_9781805120049?context=PC&vid=01PITT_INST&lang=en&adocid=cdi_walterdegruyter_marc_9781805120049&context=PC&vid=01PITT_INST&lang=en&ad) (also viewable via the [Primo discovery page](https://pitt.primo.exlibrisgroup.com/discovery/fulldisplay?docid=cdi_walterdegruyter_marc_9781805120049&context=PC&vid=01PITT_INST&lang=en&ad)  (https://pitt.primo.exlibrisgroup.com/discovery/fulldisplay?docid=cdi_walterdegruyter_marc_9781805120049&context=PC&vid=01PITT_INST&lang=en&ad)).

All required textbooks are digital and available at no additional cost through the Pitt Library and the [UPB Panther Pack](#) (https://canvas.pitt.edu/courses/381238/external_tools/11636) tool in Canvas.

Technology Requirements

- Laptop or lab desktop capable of running Unity (see Unity’s published system requirements).
- Unity Hub and the course-specified Unity version installed.
- XR Interaction Toolkit package.
- A GitHub account for version control and assignment submission.
- Access to the VR Lab (Room 113) and its headsets (Oculus Quest 2, HTC Vive Pro 2) — most class time is spent developing in VR, so attendance is essential.

Course Modality and Format

This is a face-to-face course meeting weekly in the VR Lab, with most class time spent developing directly in VR. Each class opens with a short review of prior material (5–10 minutes), roughly a half hour of discussion on the week’s topic, and lab/homework time drawn from the textbook.

Grading Policies

Exams: 10% each (30% total) | Final Project: 20% | Homework Assignments: 30% | Final Showcase: 20% | Bonuses (extra credit): 5% | Total possible: 105% (100% is the highest recorded grade).

Final course grades follow the University of Pittsburgh's official grading system and quality-point scale:

Grade	Quality Points
A+	4.00

Grade	Quality Points
A	4.00
A-	3.75
B+	3.25
B	3.00
B-	2.75
C+	2.25
C	2.00
C-	1.75
D+	1.25
D	1.00
D-	0.75
F	0.00

Communication Expectations

I will respond to emails within 24–48 hours on weekdays. Please use your Pitt email account and include the course number in the subject line.

Assignments

Weekly labs and homework are drawn from the textbook chapters listed in the Semester Schedule below, culminating in an individually chosen final VR project. Full assignment details are posted on Canvas as each unit begins.

Semester Schedule

Dates have been verified against the 2026-27 Academic Calendar. Class meets during finals week (Week 16) for the final exam.

Week	Topic	Homework
1	Intro to Course & Brainstorming Ch.1 – Introduction to XR and Unity: what VR, AR, and MR are, how they differ, and how Unity fits into XR development Interactive: Wordwall Random Picker / Wheel of Names for project ideas Lab: Instructor demo – installing Unity Hub and the Unity Editor; GitHub account setup	Write a one-page XR project concept in Markdown, push to GitHub repo.
2	The Unity Editor & Scene Creation I Ch.2 (first half) – Setting up the Unity development environment; choosing a render pipeline; touring the Editor interface, Grid and Snap Lab: Build a basic scene with primitives (ground, table, sphere) following the chapter's tutorial	Submit the basic scene with a README.md explaining the setup process.
3	The Unity Editor & Scene Creation II Ch.2 (second half) – GameObjects and components, colliders, materials, unpacking a prefab, importing from the Unity Asset Store, lighting and light/reflection probes Exam 1 (covers Ch.1–2)	Extend the scene with a custom material, an imported Asset Store object, and adjusted lighting; document in Markdown with screenshots.
4	VR Development in Unity I Ch.3 (first half) – What VR development involves; VR headset approaches; setting up a VR project with XR Plug-in Management and installing the XR Interaction Toolkit Lab: Configure a new VR project and import XR Interaction Toolkit samples	Markdown write-up of your VR project setup steps; push project + README.md.

Week	Topic	Homework
5	VR Development in Unity II Ch.3 (second half) – Exploring the XR Interaction Toolkit demo scene: the pre-configured player rig, teleporting, grabbable objects, UI elements and buttons, gaze interactions and climbing; deploying/testing with the XR Device Simulator, SteamVR, or Oculus	Document your VR interaction test (teleport + grab) in Markdown with screenshots or a short clip; push code + README.md.
6	AR Development in Unity I Ch.4 (first half) – The AR landscape; marker-based vs. markerless AR; setting up an AR project in Unity with AR Foundation; the AR Session and AR Session Origin GameObjects; placing a simple cube in AR Lab: Build a starter AR Foundation scene	Submit the AR scene with a placed object; Markdown reflection on AR Session setup.
7	AR Development in Unity II Ch.4 (second half) – Implementing plane detection and touch inputs/anchors; testing AR experiences in Unity; deploying to Android and iOS Exam 2 (covers Ch.3–4)	Add plane detection and a touch-placed object to your AR scene; push code + README.md.
8	Midterm Project Prototype Review of Ch.1–4; no new reading Lab: Each student presents a working VR or AR prototype combining scene-building, interaction, and (VR or AR) foundations from the book so far	Submit midterm prototype on GitHub with a Markdown design log.
9	Building Interactive VR Experiences Ch.5 – Interactivity without code (importing assets, teleport anchors, animator-driven rotation and scaling) and with C# (using C# to scale an object via a UI slider)	Add one no-code interaction and one C#-scripted interaction to your VR prototype; document with a UML diagram or pseudocode.

Week	Topic	Homework
10	Building Interactive AR Experiences Ch.6 – Building an interactive AR menu application: Food, ARPlacePrefab, and SwapPrefab scripts; adding UI buttons and On Click events	Implement an AR menu-style interaction (place + swap prefab) in your AR prototype; submit via GitHub with a Markdown reflection.
11	Adding Sound and Visual Effects Ch.7 – Sound theory and Unity’s audio system; building a VR drum scene with sound effects; Unity’s Particle System and scripting fog-on-collision Exam 3 (covers Ch.5–7)	Add an audio source and a particle effect to your project; push code + README.md.
12	Building Advanced XR Techniques Ch.8 – Hand-tracking with the XR Interaction Toolkit; eye- and head-gaze interaction; building a VR multiplayer application with PUN (Network Manager, avatar hands/face, animated hand models)	Prototype one advanced feature (hand-tracking, gaze interaction, or basic multiplayer connection) and document your approach in Markdown.
13	Best Practices, Future Trends & Collaboration Ch.9 – XR best practices (hardware considerations, understanding your audience, project management, UX); current and future XR trends; useful Asset Store toolkits Lab: Group activity – pair programming and peer code review via GitHub branching and pull requests	Submit revised project + Markdown “Collaboration Example” reflecting on best practices applied.
14	Testing & Polishing Lab: Students playtest peers’ prototypes and give structured feedback (Google Forms); apply Ch.9 best practices to refine UX	Submit revised project + Markdown “Playtesting Report.”
15	Final Showcase Lab: Final presentations – students demo VR/AR projects live	Prepare for final submission and exam.

Week Topic**Homework**

16

Final Exam — meets during finals week

Lab: Final Exam, online via Canvas

Submit final GitHub repo:
 Unity project files, final
 VR/AR build, Markdown
 documentation (design log,
 playtest reports, final
 reflection). Final Exam.

Other Online Tools

- Random Idea Generation / Brainstorming: [Wheel of Names](https://wheelofnames.com/) , [Random Word Generator](https://randomwordgenerator.com/) 
- Collaborative UI/UX Sketching: [Figma FigJam](https://www.figma.com/figjam/) 
- Peer Testing / Feedback: [Google Forms](https://forms.google.com) 
- Version Control & Documentation: [GitHub](https://github.com/) , [Markdown Guide](https://www.markdownguide.org/) 

University, Campus, and Course Policies

Academic Integrity

Students in this course will be expected to comply with the [University of Pittsburgh's Policy on Academic Integrity](https://www.provost.pitt.edu/academic-integrity-guidelines) . Any student suspected of violating this obligation for any reason during the semester will be required to participate in the procedural process, initiated at the instructor level, as outlined in the University Guidelines on Academic Integrity. To learn more about Academic Integrity, visit the [Academic Integrity Guide](http://pitt.libguides.com/academicintegrity/)   for an overview of the topic. For hands-on practice, complete the [Academic Integrity Modules](http://pitt.libguides.com/academicintegrity/plagiarism)  .

Extended Academic Integrity Statement

Members of the University community, both faculty and students, bear a serious responsibility to uphold personal and professional integrity and to maintain complete honesty in all academic work. Violations of the code of academic integrity are not tolerated. Students who cheat or plagiarize or who otherwise take improper advantage of the work of others face harsh penalties, including permanent dismissal. Incidents of forged signatures that are associated with any academic endeavor at Pitt-Bradford, in addition to being a criminal offense, are viewed as violations of academic integrity. The academic integrity guidelines set forth student and faculty obligations and the means of enforcing regulations and addressing grievances. Violations of academic integrity will be tracked by the Office of Academic Affairs. Refer to the [Pitt-Bradford Academic Procedures & Policies](https://catalog.upb.pitt.edu/content.php?catoid=231&navoid=24277)  for general guidelines on academic integrity. Copies of the complete

Guidelines on Academic Integrity are available [here \(https://www.provost.pitt.edu/faculty/academic-integrity-freedom/academic-integrity-guidelines\)](https://www.provost.pitt.edu/faculty/academic-integrity-freedom/academic-integrity-guidelines).

Disability Services

The University of Pittsburgh at Bradford is committed to inclusive and accessible education experiences for all students. If you have a disability for which you are, or may be, requesting an accommodation, you are encouraged to contact the Coordinator for Equity and Accessibility as early as possible in the term: Chad Hockaday, [cjh187@pitt.edu \(mailto:cjh187@pitt.edu\)](mailto:cjh187@pitt.edu), 814-362-7609. Equity and Accessibility staff will verify your disability and determine reasonable accommodations for this course.

Accessibility

The University of Pittsburgh is committed to [providing equal access \(https://www.wellbeing.pitt.edu/disability-access-accommodations/digital-accessibility/policy\)](https://www.wellbeing.pitt.edu/disability-access-accommodations/digital-accessibility/policy) by making all electronic and information technology accessible to all users. This includes instructional materials and the [Canvas LMS platform ↗ \(https://community.instructure.com/en/kb/articles/662723-what-are-the-canvas-accessibility-standards\)](https://community.instructure.com/en/kb/articles/662723-what-are-the-canvas-accessibility-standards).

- If you experience accessibility issues regarding course-specific material, please reach out to me, your instructor.
- If you experience a University-wide or other large-scale accessibility issue, please [report the digital accessibility barrier \(https://www.wellbeing.pitt.edu/disability-access-accommodations/digital-accessibility/report-a-barrier\)](https://www.wellbeing.pitt.edu/disability-access-accommodations/digital-accessibility/report-a-barrier) to the Digital Accessibility Coordinator.
- If you are registered with Equity and Accessibility and experience issues with approved accommodations, you may contact Chad Hockaday, [cjh187@pitt.edu \(mailto:cjh187@pitt.edu\)](mailto:cjh187@pitt.edu), 814-362-7609.

Religious Observances

The University of Pittsburgh at Bradford strives to foster a supportive environment and to respect the beliefs and values of all community members. Accordingly, recognizing that certain students may observe religious holidays and cultural practices that conflict with scheduled class activities, I will work with you to try to make reasonable accommodations in the event of such a conflict.

At the beginning of the semester, you should review the course requirements to identify foreseeable conflicts with assignments, exams, or other required attendance. If you identify a conflict, please contact me, ideally within the first two weeks of the first class meeting, to allow time to discuss and attempt to make fair and reasonable adjustments to the schedule and/or tasks.

Copyright Notice

Course materials may be protected by copyright. United States copyright law, 17 USC section 101, et seq., in addition to University policy and procedures, prohibit unauthorized duplication or retransmission of course materials. For more, see the [Library of Congress Copyright Office ↗](#)

(<https://www.copyright.gov/>) and the [University Copyright Policy](#)  (<https://pitt.libguides.com/copyright/pittpolicies>).

Statement on Classroom Recording

To ensure the free and open discussion of ideas, students may not record classroom lectures, discussion, and/or activities without the advance written permission of the instructor or without explicit permission from the Office of Equity and Accessibility as a result of registered accessibility needs. Any such recording properly approved in advance can be used solely for the student's own private use.

The University of Pittsburgh at Bradford has contracted with Panopto to provide lecture capture services. The VR Lab may be equipped to record computer screen, classroom audio, and video. Instructors are not required to use Panopto. Any student not wishing to be recorded must inform their instructor in writing at the earliest possible convenience. Recorded lectures are provided for student use only and are the property of the instructor and the University of Pittsburgh.

Bias, Discrimination, and Harassment Reporting

The faculty and staff at Pitt-Bradford are committed to supporting our students and seeking an environment that is free of bias, discrimination, and harassment. If you have encountered any form of bias, discrimination, or sexual misconduct (e.g., sexual assault, sexual harassment, stalking, domestic or dating violence), I encourage you to report this to the University. If you speak with a faculty or staff member about an incident of misconduct, that employee will inform you if they are mandated by law to notify one of the campus's Title IX/Affirmative Action liaisons regarding the incident. Students may contact Title IX liaison [Dr. Drew Peterson \(mailto:dtp46@pitt.edu\)](mailto:dtp46@pitt.edu) (814-362-7651), Vice President and Dean of Student Affairs, located on the second floor of the Frame-Westerberg Commons. If you wish to speak with a confidential source, contact campus [Counseling Services \(https://www.upb.pitt.edu/life-at-pitt-bradford/health-and-wellness/counseling-services\)](https://www.upb.pitt.edu/life-at-pitt-bradford/health-and-wellness/counseling-services) at 814-362-5272.

Generative AI Use Statement

Some Use of Generative AI Permitted Under Some Circumstances or With Explicit Permission: During this class, we may use Generative AI tools such as [Claude \(https://services.pitt.edu/TDCClient/33/Portal/Requests/Service/397/Claude-for-Education\)](https://services.pitt.edu/TDCClient/33/Portal/Requests/Service/397/Claude-for-Education). You will be informed as to when, where, and how these tools are permitted to be used, along with guidance for attribution. Any use outside of this permission constitutes a violation of the University of Pittsburgh at Bradford's [Guideline on Academic Integrity \(https://catalog.upb.pitt.edu/content.php?catoid=237&navoid=26354#academic-integrity\)](https://catalog.upb.pitt.edu/content.php?catoid=237&navoid=26354#academic-integrity).

Plagiarism Statement

Plagiarism is using another person's words and/or ideas as if they were your own. It is easy to avoid plagiarism; simply put the language you have taken from someone else in quotation marks and cite the person's name and source in your paper. If you are using another thinker's idea but not necessarily their words, include a citation to indicate who came up with it and where you found it.

Plagiarism is extremely serious in every instance. If done purposefully, it is nothing less than sheer academic dishonesty; if it occurs by accident, plagiarism demonstrates a critical lack of understanding about the usage of source materials that must be addressed immediately.

It is your responsibility as an ethical member of our academic community—and as an adult—to understand what plagiarism is, to avoid it at all costs, and to recognize its detrimental impact upon your development as a writer and as a human being. You owe it to yourself to check out the [Pitt Academic Procedures & Policies \(https://www.provost.pitt.edu/academic-integrity-guidelines\)](https://www.provost.pitt.edu/academic-integrity-guidelines) “Guidelines on Academic Integrity.”

Teaching Survey Statement

Students in this class will be asked to complete a [Student Opinion of Teaching Survey \(https://teaching.pitt.edu/omet/student-information/\)](https://teaching.pitt.edu/omet/student-information/). Surveys will be sent via Pitt email and appear on your Canvas landing page during the last three weeks of class meetings. Responses are confidential, and I don't see results until after final grades are posted. Please take time to thoughtfully respond; your feedback is important to me.

Final Exam Policy

Students with more than two final examinations scheduled for the same day (excluding evening classes) may petition their instructors to assist in rescheduling one exam at a time mutually agreeable to the student and instructor. If it is not possible to reach an agreement with one of the instructors, students should contact the appropriate division chairs.

Online tutorials for Canvas are available at:

<https://community.canvaslms.com/community/answers/guides/> 

[\(https://community.canvaslms.com/community/answers/guides/\)](https://community.canvaslms.com/community/answers/guides/)

Severe Weather Statement

Pitt-Bradford's general severe weather policy is to remain open in all but the most extreme circumstances. Individuals must make their own decision whether or not to come to class or report for work. Faculty are expected to be understanding when students miss class because of severe weather and to not penalize students who are unable to travel to campus due to severe weather conditions. All university faculty, staff, and students are urged to use their own discretion in deciding whether they can safely commute to class or to work.

Classroom Community and Civility Statement

Every student brings to the classroom a unique point of view. We tend to think and learn in our own way, based in part on our own social and cultural background. In this course, we will create an environment in which people of all identities are encouraged to contribute their perspectives to academic discourse. Come to class prepared to conduct yourself with civility, respect, and integrity, and help create a safe space for discussions that are intellectually rigorous and mature.

- Use “I” statements — you can only speak from your personal experiences, not as a representative of a group.
- Respect other students’ preferences regarding the name and pronouns by which they would like to be referred and strive to utilize inclusive language.
- Do not be judgmental regarding others’ opinions and perspectives.
- Approach the course content in an academic, scholarly manner.
- Create an atmosphere of mutual respect and be considerate of others’ feelings.
- Work to develop an understanding of different points of view.
- Have an open mind and reserve the right to change your mind.
- Be patient with each other and yourself.

Class Cancellation Procedures

There may be times during the semester that a class needs to be canceled. Here's what we'll do to keep you notified:

- When possible, I will email students directly informing you of the cancellation.
- If I am unable to email students, the division administrative assistant will email you.
- The division administrative assistant will also post a notice on the classroom door.

Please bear in mind that sometimes emergencies or unforeseen circumstances may prevent timely notification — but all possible efforts will be made to ensure that you are informed in a timely manner.

Please review the most up-to-date policies and procedures in the [Annual Security and Fire Safety Report](https://www.upb.pitt.edu/sites/default/files/files/2021-06/security-brochure_Campus-Police.pdf) (https://www.upb.pitt.edu/sites/default/files/files/2021-06/security-brochure_Campus-Police.pdf).

Some things you can do to make sure you know when a class has been canceled:

- Check your Pitt email before you leave for class.
- Check Canvas for any possible notifications.
- Make sure your contact information is up-to-date in [edu \(https://my.pitt.edu/\)](https://my.pitt.edu/).
- Register for Pitt's [Emergency Notification Service](http://www.technology.pitt.edu/portal/emergency/emergency-notification.html) (<http://www.technology.pitt.edu/portal/emergency/emergency-notification.html>).

Contact a division administrative assistant if you suspect your class may be canceled:

- [Behavioral & Social Sciences \(https://www.upb.pitt.edu/academics/academic-divisions/behavioral-and-social-sciences\)](https://www.upb.pitt.edu/academics/academic-divisions/behavioral-and-social-sciences)— Lindsey Brandon, 362-7620
- [Biological & Health Sciences \(https://www.upb.pitt.edu/academics/academic-divisions/biological-and-health-sciences\)](https://www.upb.pitt.edu/academics/academic-divisions/biological-and-health-sciences)— Laurie Dennis, 362-7640
- [Communication & the Arts \(https://www.upb.pitt.edu/academics/academic-divisions/communication-and-arts\)](https://www.upb.pitt.edu/academics/academic-divisions/communication-and-arts)— Shelly Whitman, 362-7590
- [Management & Education \(https://www.upb.pitt.edu/academics/academic-divisions/management-and-education\)](https://www.upb.pitt.edu/academics/academic-divisions/management-and-education)— Lindsey Brandon, 362-7620

- [Physical & Computational Sciences \(https://www.upb.pitt.edu/academics/academic-divisions/physical-and-computational-sciences\)](https://www.upb.pitt.edu/academics/academic-divisions/physical-and-computational-sciences) – .lanet Shade 362-7560

Course Summary:

Date	Details	Due
Thu Nov 14, 2024	 Milestone 1 - Draft Scene (https://canvas.pitt.edu/courses/381238/assignments/2244238)	due by 11:59pm
	 Technical Project Definition (https://canvas.pitt.edu/courses/381238/assignments/2244240)	due by 11:59pm
Sun Sep 28, 2025	 Alumni Weekend Credit (https://canvas.pitt.edu/courses/381238/assignments/2244234)	due by 11:59pm
Wed Oct 1, 2025	 Unity Hello World (https://canvas.pitt.edu/courses/381238/assignments/2244241)	due by 11:59pm
Thu Oct 2, 2025	 VR GameObjects & Components (https://canvas.pitt.edu/courses/381238/assignments/2244243)	due by 11:59pm
Thu Oct 9, 2025	 Exam 1 (https://canvas.pitt.edu/courses/381238/assignments/2244233)	due by 11:59pm
Thu Oct 30, 2025	 Headset + Building Blocks (https://canvas.pitt.edu/courses/381238/assignments/2244237)	due by 11:59pm
Thu Nov 20, 2025	 Exam 2 (https://canvas.pitt.edu/courses/381238/assignments/2244232)	due by 11:59pm
Sun Dec 14, 2025	 Complete VR Project (https://canvas.pitt.edu/courses/381238/assignments/2244235)	due by 11:59pm
	 Final Exam (https://canvas.pitt.edu/courses/381238/assignments/2244231)	due by 11:59pm
Wed Sep 2, 2026	 GitHub Profile Link (https://canvas.pitt.edu/courses/381238/assignments/2244236)	due by 11:59pm
Thu Sep 24, 2026	 VR Concept (https://canvas.pitt.edu/courses/381238/assignments/2244242)	due by 11:59pm

Date	Details	Due
Tue Sep 29, 2026	 Campaign Launch Party - Extra Credit (https://canvas.pitt.edu/courses/381238/assignments/2276329)	due by 11:59pm
Wed Sep 30, 2026	 Alumni Weekend - Virtual Reality Lab Experience - Extra Credit (https://canvas.pitt.edu/courses/381238/assignments/2276331)	due by 11:59pm
	 PYURA Friday 11/14/2025 (https://canvas.pitt.edu/courses/381238/assignments/2244239)	